CURIONEST

Chapter 1 — Math & Measurement

Unit Pack · Measurement · Sig Figs · Density · Scientific Notation · Dimensional Analysis

WHAT'S INSIDE

✓ Complete concept notes with worked examples

✓ Practice sets in varied formats (short answer, MCQ, calculations)

✓ Key terms glossary + full answer key with worked solutions

✓ Original figures and Word-native equations — print-ready & editable

Made by a chemistry teacher — aligned to the standard US chemistry sequence.

In this unit, you will learn to:

1. Distinguish qualitative and quantitative observations
2. Identify SI base units and metric prefixes
3. Count significant figures and round calculations correctly
4. Calculate density using numerical set-ups
5. Convert between standard and scientific notation
6. Convert units using dimensional analysis

Unit Vocabulary — Write each definition in your own words

Word	Definition (your own words)
accuracy	
precision	
conversion factor	
density	
mass	
qualitative observation	
quantitative observation	
scientific notation	
significant figures	
unit	

LESSON 1: MEASUREMENT

Objective:

- Distinguish qualitative from quantitative observations
- Identify SI base units and metric prefixes

Chemistry is the study of matter and its changes. Matter can be described through its properties using two kinds of observations. Qualitative observations are descriptive and non-numerical (color, odor, texture). Quantitative observations are in the form of numbers and units (mass 5 g, temperature 371 K).

Every measurement has three parts: a number (magnitude), a unit (standard of comparison), and an indication of uncertainty. The SI system uses base units: meter (length), kilogram (mass), second (time), kelvin (temperature). Metric prefixes change the size of a unit by powers of ten: kilo (k, 1000×), centi (c, 1/100), milli (m, 1/1000), micro (μ, 1/1,000,000).

Worked Example

Classify each observation as qualitative or quantitative: (a) the metal is silver-colored; (b) it melts at 371 K; (c) its density is 0.97 g/cm³.

Step 1: Silver-colored describes appearance without numbers → qualitative.

Step 2: 371 K is a number with a unit → quantitative.

Step 3: 0.97 g/cm³ is a number with units → quantitative.

Answer: (a) qualitative, (b) quantitative, (c) quantitative.

Check Your Understanding

1. Which is a QUALITATIVE observation?

(A) The liquid is clear
(B) The mass is 45.8 g
(C) The temperature is 298 K
(D) The volume is 25.0 mL

Answer: _______

2. Which SI base unit measures temperature?

(A) Celsius
(B) Kelvin
(C) Fahrenheit
(D) Joule

Answer: _______